Presentation Script — End-Semester Review

Deepfake Detection on FaceForensics++ · 23AID205 · 14 slides, target 15:00 (budgeted 14:50)

#	Presenter	Slides	Starts	Budget
1	Vipin Sudhakar	1–4	0:00	3:20
2	Rithvik Arulprakash	5–7	3:20	3:30
3	Venugopalan Gangadharan	8–10	6:50	4:15
4	Harshith Kv	11–14	11:05	3:45

Slide 1 (title) is opened by Vipin, who continues straight into slides 2–4.

Vipin Sudhakar

Starts at 0:00 — 3:20 budget

Slide 1 Title — 0:20

• Good [morning/afternoon]. We're presenting our end-semester review for 23AID205: deepfake detection on FaceForensics++.

• We started by comparing two detectors, and ended up studying why detectors fail on fakes they haven't seen. I'm Vipin, and I'll start.

Slide 2 Motivation, Problem & Objectives — 1:00

• Deepfakes are a real fraud and misinformation risk. Detectors are usually tested on the same kind of fake they were trained on.

• That hides the real problem: a detector at 99% can fail on a new kind of fake, and nobody notices until it's deployed.

• Our first objective was the course comparison: XceptionNet vs. a CNN-ViT hybrid. Our second came from our own results: explain why some scores fell below chance.

Slide 3 End-to-End Pipeline — 1:00

• Six steps, left to right: raw videos, face crops, official splits, training, evaluation, analysis.

• Splits are by video, not frame, so the same person is never in both train and test.

• We train on one manipulation type and test on all four. Every number comes from a script over saved scores, not typed by hand.

Slide 4 Dataset and Model Selection — 1:00

• FaceForensics++: 1,000 real YouTube videos and 4,000 fakes from four methods.

• We used a Kaggle copy because official access is still pending, so we label results as preliminary. We checked it for problems before training.

• XceptionNet vs. the hybrid is the course comparison. ResNet-50 and EfficientNet-B4 came later, to test whether the failure depends on the model.

Handoff: Rithvik will take you through the results.

Rithvik Arulprakash

Starts at 3:20 — 3:30 budget

Slide 5 Course Experiments: XceptionNet vs. CNN-ViT 1:15

• Same-method, XceptionNet reached AUC 0.996 and the hybrid 0.984.

• On unseen methods both dropped to about 0.60 and 0.57. The hybrid did not generalize better, a negative result we report honestly.

• Under heavier compression both degrade a lot. We also built a live web demo that runs both models.

Slide 6 Key Finding: Below-Chance Transfers 1:30

• Here's the surprise. Train on Deepfakes, test on FaceSwap: the AUC is 0.26, below 0.5.

• Below 0.5 means the detector ranks the fakes as more real than real faces: it is confidently backwards, not just unsure.

• We rebuilt the whole preprocessing pipeline, with a different face detector, alignment, format and size, and got 0.26 again. So it's not a pipeline bug.

Slide 7 Grad-CAM: What the Models Look At 0:45

• Same model, same video. On Deepfakes it looks at the face. On FaceSwap its attention collapses off the face, and it calls the fake real with probability near zero.

Handoff: Venu will explain why this happens.

Venugopalan Gangadharan

Starts at 6:50 — 4:15 budget

Slide 8 No Single Preprocessing Change Moves DF → FS 1:15

• We changed one preprocessing step at a time: JPEG quality, number of frames, and how tightly we crop the face.

• Deepfakes to FaceSwap stayed at about 0.26 every time. It's the most robust result in the study.

• We wrote down predictions before seeing these results. Two held, two didn't, and we report both.

Slide 9 A Sharpness Cue Explains Which Cells Invert 1:45

• The cause: Deepfakes fakes are blurrier than real faces (d = -0.48), and FaceSwap fakes are sharper (d = +0.37).

• A detector trained on Deepfakes learns 'blurry means fake'. FaceSwap fakes are sharp, so it calls them real. That's the inversion.

• This one rule predicts our whole matrix (rank correlation 0.92) and a published result from another lab on the official data (0.73).

Slide 10 Randomizing Sharpness Partly Fixes It 1:15

• To test cause and effect, we randomly blurred or sharpened every training image, so sharpness stopped being useful.

• Deepfakes to FaceSwap rose by +0.10 and FaceSwap to NeuralTextures by +0.09, with no loss on the detector's own task.

• But Deepfakes to FaceSwap is still 0.36, below chance. So sharpness is a real cause, but not the whole story.

Handoff: Harshith will cover how far this generalizes and wrap up.

Harshith Kv

Starts at 11:05 — 3:45 budget

Slide 11 Same Failure Across Architectures, Not Across Datasets 1:15

• Across architectures it holds: ResNet-50 and EfficientNet-B4 fail exactly like Xception, in all 8 cells. So it comes from the training data, not the network.

• Across datasets it doesn't: on DeepFakeFace the sharpness rule was right in 2 of 4 cells, which is chance. Our explanation is specific to FaceForensics++.

Slide 12 Verdict on the Four Claims 1:00

• Preprocessing causes it: not supported. The mechanism: supported within FaceForensics++. The fix: partial. Architecture independence: supported.

Slide 13 Why This Could Be Publishable 1:00

• What's new: we explain which transfers invert, with one measurable cue that also predicts another lab's published results.

• Why it's credible: predictions recorded in advance, three seeds, confidence intervals, and all code public.

• We'd target a workshop. The risks are the unofficial data, the partial fix and the second dataset, and we state all three openly.

Slide 14 Limitations, Future Work & Team 0:30

• Next: re-run on official data and find what still drives the remaining inversion. A paper draft is ready. Thank you, we're happy to take questions.

Likely Questions

Short answers, with who should take each one.

Isn't an AUC below 0.5 just a bug, like flipped labels?

We checked: the audit's polarity check confirms fakes are labelled 1 and the AUC can't silently flip, the audit passed 17/17 checks, and a completely rebuilt pipeline gives the same number. A published paper also reports this cell at chance on official data. (Venu / Rithvik)

Why didn't the CNN-ViT hybrid do better?

It was trained from scratch at 224 px while Xception was pretrained at 299 px, so the comparison isn't fully fair; we say so. The later study shows the failure doesn't depend on the architecture anyway. (Rithvik)

Why a Kaggle copy and not the official dataset?

Official access was requested but is still pending. Our sharpness rule also predicts a published official-data result, which suggests the mirror isn't the cause. (Vipin)

Can you explain the code?

Open docs/project_code_simplified.py and go top to bottom: splits (1), face cropping (2), dataset and the blur/sharpen augmentation (3), the two models (4), training (5), evaluation with confidence intervals (6), the cross-manipulation matrix (7) and the sharpness rule (8). Each section names the real file it comes from. (any presenter)

Is it publishable?

As a workshop or short paper, yes: the cue, its prediction of published results and the intervention are new. Main-track would need official data and a full fix. (Harshith)